

Full-Stack On-Site Programming Assignment

FSC Code Classifier

Overview

Every day, SalesPatriot scrapes over **4,000 government solicitations** valued at more than **\$160 million**. That's a lot of opportunities - but not every contract is relevant to every company.

Consider two of our customers: Company A manufactures light bulbs, while Company B produces aircraft landing gear. Clearly, they're interested in very different contracts. Our job is to filter away the noise and surface only the opportunities that matter to each customer.

How do we do this? The U.S. federal government classifies every procurement using **Federal Supply Classification (FSC)** codes - 4-digit codes that categorize products and services. If we know which FSC codes apply to a company, our matching engine can automatically connect them to relevant solicitations.

That's where you come in. Your task is to build a tool that takes company information as input and outputs the FSC codes that best describe what they sell or provide.

This is an open-ended problem with no single "correct" solution. We want to see how you think, how you approach an unfamiliar challenge, and how you make tradeoffs under time constraints.

The Challenge

Build it your way. Choose your own tech stack, architecture, and approach. Create a full-stack web application where users can:

1. **Upload/input company information** including:
 - Company name
 - Website URL
 - Email domain
 - Any other relevant documentation
2. **Process the information** to extract meaningful features about what the company does, sells, or provides.
3. **Display recommended FSC codes.** The only requirement is outputting the **relevant** 4-digit codes - everything else is up to you.

Resources

FSC Code Reference: The complete FSC code list is publicly available at https://www.dla.mil/Portals/104/Documents/Aviation/AviationEngineering/Engineering/AV_FSCClassAssignment._151007.pdf.

API Keys: We will provide API keys for any paid services you need. Just let us know what you'd like to use (LLMs, OCR, vector databases, etc.).

Deliverable

Push your code to a Git repository and share the link with us. Be ready to give a short demo of your solution and walk us through your approach - we value out of the box thinking and creativity.

Test Data

You'll receive a zip file with 3 sample companies to test your solution against. Use these to validate your approach. We will also provide a fourth company for you to test your solution against during the demo.

Time

You have **a few hours** for this assignment. This is about **rapid prototyping**, not production-ready code. A working MVP that shows your problem-solving approach is far more valuable than a polished but incomplete solution. Don't get stuck on perfection - show us how you think.

What We're Looking For

- **Problem-solving** - How do you break down an ambiguous problem?
- **Creativity** - What's your unique approach to the challenge?
- **Prototyping speed** - Can you build something functional under constraints?
- **Communication** - Can you explain your decisions and tradeoffs?

Questions?

If you need clarification or want to request API keys, just ask us.